ExplainableCropAI: Multi-level Explanation Process

Input Image

Field-captured paddy leaf RGB 640 × 640 × 3

RGB image

ConvNeXt V2-Tiny NOVEL

Modern CNN backbone for feature extraction

Multi-scale features (P2–P5)

Top features (P5)

Visual Explanation NOVEL

Grad-CAM++ attribution Second-order gradients produce heatmaps over disease-relevant regions

Uncertainty

Monte Carlo Dropout 30 stochastic passes split into epistemic and aleatoric components

Visual evidence

Prediction + uncertainty

Phi-3-mini LLM NOVEL

Audience-adaptive natural-language generation fed with prediction, uncertainty and visual evidence

Diagnosis context

Recommendation Engine

Context-aware treatment guidance, conditioned on diagnosis, uncertainty and agronomic priors

Agronomic rules

Knowledge Base

Agronomic rules, treatments, severity thresholds

Explanation

Treatment

Decision Support Output

Three synchronised channels:
prediction + confidence | visual explanation | NL explanation + recommendations

Deployed insights

End User / Mobile App

Farmer or extension officer

Components

- Input / Output
- CNN Backbone
- Visual Explanation
- Uncertainty
- LLM Module
- Recommender
- Knowledge Base
- N — Novel contribution